

Nexus State Warehouse

A scientific data warehouse for structural, kinetic, and thermodynamic inference

CALYRAI × NEXUS

Structural biology / SAXS / SPR / cryo-EM / G-space inference



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states / warehouse / hypothesis generation

Problem

Scientific data is fragmented:

- ▶ SAXS $\rightarrow$ structure: $p(r)$, R_g , D_{max}
- ▶ SPR $\rightarrow$ kinetics: k_a , k_d
- ▶ Cryo-EM $\rightarrow$ geometry: surface and curvature
- ▶ PCA $\rightarrow$ thermodynamics: G-space and low-dimensional state maps

Current failure mode:

- ▶ no unified query layer
- ▶ no shared state representation
- ▶ no cross-domain inference object

Vision

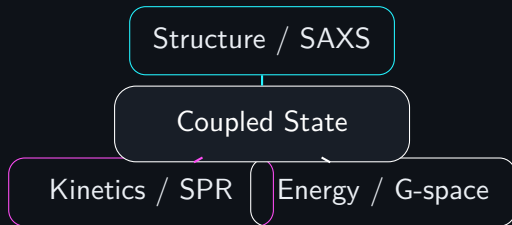
Replace datasets with **states**

State = minimal physically consistent description of a system

Instead of querying experiments, Nexus queries biological and physical states that multiple experiments constrain together.

Core Concept

- ▶ Structure → SAXS
- ▶ Geometry → Cryo-EM
- ▶ Kinetics → SPR
- ▶ Energetics → PCA / G-space



Architecture

RAW DATA → FEATURES → STATES → WAREHOUSE →
HYPOTHESES

The warehouse is the middle layer where heterogeneous experimental evidence becomes one queryable state object.

Data Model

- ▶ Dataset
- ▶ Feature
- ▶ State
- ▶ Model
- ▶ Hypothesis

These are not business tables. They are scientific abstractions for linking observables to admissible physical states.

SQL Schema

```
CREATE TABLE datasets (  
  dataset_id TEXT PRIMARY KEY,  
  method TEXT,  
  path TEXT,  
  temperature FLOAT,  
  ph FLOAT  
);
```

```
CREATE TABLE features (  
  feature_id TEXT PRIMARY KEY,  
  dataset_id TEXT,  
  name TEXT,  
  value FLOAT,  
  unit TEXT  
);
```

```
CREATE TABLE states (  
  state_id TEXT PRIMARY KEY,  
  system TEXT,  
  label TEXT,  
  confidence FLOAT
```

Scientific Query Example

Question: Does structural expansion correlate with stronger binding?

```
SELECT s.state_id
FROM states s
JOIN state_features sf1 ON s.state_id = sf1.state_id
JOIN features f1 ON sf1.feature_id = f1.feature_id
JOIN state_features sf2 ON s.state_id = sf2.state_id
JOIN features f2 ON sf2.feature_id = f2.feature_id
WHERE f1.name = 'Rg' AND f1.value > 5.0
      AND f2.name = 'kd' AND f2.value < 0.002;
```

Hypothesis Layer

Example state:

- ▶ R_g increases
- ▶ k_d decreases

Hypothesis:

- ▶ structural expansion stabilizes binding
- ▶ confidence is derived from multi-modal agreement

Why This is Different

Classical data warehouse

- ▶ business metrics
- ▶ static tables
- ▶ dashboards

Nexus warehouse

- ▶ physical states
- ▶ cross-domain coupling
- ▶ hypothesis generation

Modern Stack

Core:

- ▶ DuckDB for local analytical execution
- ▶ Apache Arrow for columnar transport
- ▶ Python for orchestration and scientific execution

Extensions:

- ▶ graph database for state relations
- ▶ feature store for ML integration
- ▶ versioned object storage for large experimental payloads

Alternative Architectures

Graph-based:

- ▶ direct state-to-feature and state-to-state relations
- ▶ natural representation of coupled systems

Lakehouse:

- ▶ versioned datasets
- ▶ time-travel queries
- ▶ reproducible warehouse snapshots

Your Edge

AI learns correlations

Nexus learns physics

Conclusion

Nexus State Warehouse enables querying biological reality